

Survey of System-on-Chip Modular Test Approach

Harpreet Vohra^{1,*}, Amardeep Singh²

¹Department of Electronics and Communication Engineering, Thapar University,
Patiala, Punjab, India

²Department of Computer Science and Engineering, Punjabi University, Patiala, Punjab, India

Abstract

The advancement in semiconductor technology over the past few decades has enabled the industry to develop new design and reuse methodologies referred as System-on-chip (SoC) design. These typically integrate heterogeneous mix of various complex analog and digital cores, with all their necessary electronic circuitry. The recent advancements have led to the use of the 3D structures (3D-SOCs) that brought along the benefits of higher performance, minimum average interconnect length, reduced power and smaller footprint. These revolutions have not only brought in new issues of design but also their testing which is becoming the major bottleneck in determining the overall system cost and a defect free delivery to market in time. To facilitate the testing of cores lying inside such complex SOC's modular approach, concurrency in test application are the only viable ways. However these come at their own costs of constraints like power, test bandwidth, resources like "through silicon vias", BIST etc. This paper first discusses in general the various challenges in testing of core-based system chips, the associated test factors and modular test approach. It is further followed up by the various ways of addressing the challenges and the corresponding research areas.

Keywords: System on chip test, test data compression, test access architecture, embedded core based design, system on chip test architecture

***Author for Correspondence** E-mail: vohra.harpreet@yahoo.com

INTRODUCTION

To cater to the ever-growing demands of the consumer market, the VLSI industry is being forced to add more and more complexity on chip. The achievement of the remarkably high level of integration has been made possible due to the corresponding advancements in the semiconductor industry. Due to this, it is now common to find complete systems-on-chip (SOC) consisting of a mix of analog/ digital blocks in the form of integration of predefined and pre verified embedded cores. Embedded core based design approach on one hand saves the design cost while on the other it leads to a strong benefit of getting the various IP blocks from the vendors who are stalwarts in their respective domains. These IP cores can be in the form of a synthesizable HDL code, a gate level netlist or a layout description of a circuit. It has been observed that though miniaturization in the component sizes has been achieved however, the interconnects have not scaled down at the same pace at the global level and even if they have been at the local

interconnect level, the R-L-C component have led to performance issues [1]. To cope up with the demand for higher integration 3D SOC [2] has started to gain considerable attention that allows growth in the vertical dimension. Big designs with smaller foot prints can be obtained by stacking multiple dies together.

The ITRS roadmap predicts 3D integration as a key technique to overcome this so-called "wiring crisis" [3]. 3D ICs employs through silicon via (TSV) for the interconnectivity. This fast growing advancement in the technology has led to the challenges for the design engineer and a more complicated and expensive problem for test engineers.

The importance of reducing the cost of test is further motivated by comparing the test cost with the cost of fabrication. Today, the cost of test is a significant part of the overall manufacturing cost (including the cost of fabrication and the cost of test). All this makes it mandatory to have a good and reliable

system test plan as early as possible to save the end cost. There are several challenges that the test integrator faces when designing the test strategies for a core based SOC. Since large numbers of cores are integrated on chip then to ensure their productivity, the SOC has to be tested for various types of faults associated with the cores. This leads to test generation and analysis issues like the decision on how the test vectors have to be generated (i.e. externally with the help of an Automatic test pattern generator, stored internally in some ROM or generated internally using some built in test circuitry). Similarly for the response analysis one has to decide on the use of off chip or on chip signature analyzers.

Test Challenges

There are several challenges that the test integrator faces when designing the test strategies for a core based SoC, these are as discussed below:

- Since large numbers of cores are integrated on chip then to ensure their productivity, the SOC has to be tested for various types faults associated with the cores. This lead to test generation and analysis issues like a decision on how the test vectors have to be generated (i.e. externally with the help of an ATPG, stored internally in some ROM or generated internally using some built in test circuitry). Similarly for the response analysis one has to decide on the use of off chip or on chip signature analysers.
- System integration also aggravates the problem of controllability and observability. In order to address this problem the test engineer has to decide about the approaches such as scan point insertion which for the bigger designs is becoming infeasible or design for testability which further increases the area overhead.
- Large number of cores with large fault lists associated with them lead to huge test data volume which needs to be generated and stored somewhere leading to large memory requirements.
- Large test data volume requires large test application time and hence SoC integrator has to decide on the Test Access Mechanism and scheduling in case of concurrent testing.
- At speed testing, non-correlation between the test data, consecutive activation of different modules of the design, serial shifting of the test data for multiple clocks in scan based testing leads to drastic increase in switching activity which further leads to huge power consumption.

The high test cost for core-based SOC's can be reduced by adequate test planning.

SOC TEST ARCHITECTURE

In order to cover all the possible faults which are likely to be present in an IC as a part of the manufacturing defects, test vectors need to be generated and applied to the cores. For the checks the response also need to be verified against that produced by the good machine (one that meets the desired specification). As the complexity has increased the accessibility of various cores and their associated terminals become difficult to access. In order to ensure the testability of the complete design one has to either insert the special test points or rely on structural design for testability techniques. Zorian [4] proposed the modular testing approach which requires isolating the cores with respect to the other. A conceptual architecture for testing embedded core-based SOC is shown in the Figure 1. The various hardware components can be broadly classified as:

- Test vector source and sink
- Test access mechanism
- Core test wrappers

Automatic test equipment (ATE) performs dual function: It acts as the source of the test vectors and sink for the various results obtained after application of the test data to the circuit under test. Herein, various test vectors corresponding to the various possible faults are saved in the available ATE memory chip in the compressed /uncompressed form (explained later). The test data is further forwarded to the SOC periphery through the available SOC test pins. The compressed test data is uncompressed to regain the original form with the help of an on chip decompressors are further applied to it through the test access mechanism. Test Access mechanism (TAM) works as the carrier of the test vectors to the various on chip cores. The cores assignment and their schedules of test put a huge impact

on the test application time and power. Therefore, an efficient test access architecture design turns up as an integral part of the test plan development of SOC.

Once the test vectors reach their associated cores, they are applied to the circuit under test through the test wrapper. The core test wrapper works as an interface between the core under test and the TAM. It works on deserialization of the test data (as received from the TAM) and its application to the respective core terminals (functional input/output and internal scan chains).

For a particular circuit the test pattern generation can be either exhaustive, random, semi random. This can be done by inserting an on chip hardware (as shown to be associated with RAM in Figure 1) called Built In self - test (BIST) [5, 6] that through the help of linear feedback shift registers (LFSR) allows the on chip test pattern generation and can also help observing the responses using the MISRs. BIST simplifies the task of the system test engineer as it only requires to set the BIST mode for test Vector generation. However, this demands an area and performance overhead. Most of the IP cores are normally not BIST

ready and due to the increase in the integration level with technology it becomes almost compulsory to rely on other structural techniques to keep record of all the test patterns to achieve the desired fault coverage.

The test patterns can also be generated through software employing special automatic test generation algorithms (ATPG) and stored externally in the memory of a machine termed as automatic test equipment (ATE). ATE acts as the source (Figure 1) of the test patterns from where these are supplied to the CUT similarly the responses of the circuit under test can be brought outside using the Test Access Mechanism (TAM) to a sink present in ATE where it is compared with the golden machine responses held in the memory.

The continuous up gradation in the integration, technology, architecture and performance necessitates the installation of an ATE with features such as huge memory depths, high-frequency channels, high pin count, as well as special features for analog and RF cores is expensive. All this demands to have ATE technology advancements simultaneous with the changing trend in the IC industry.

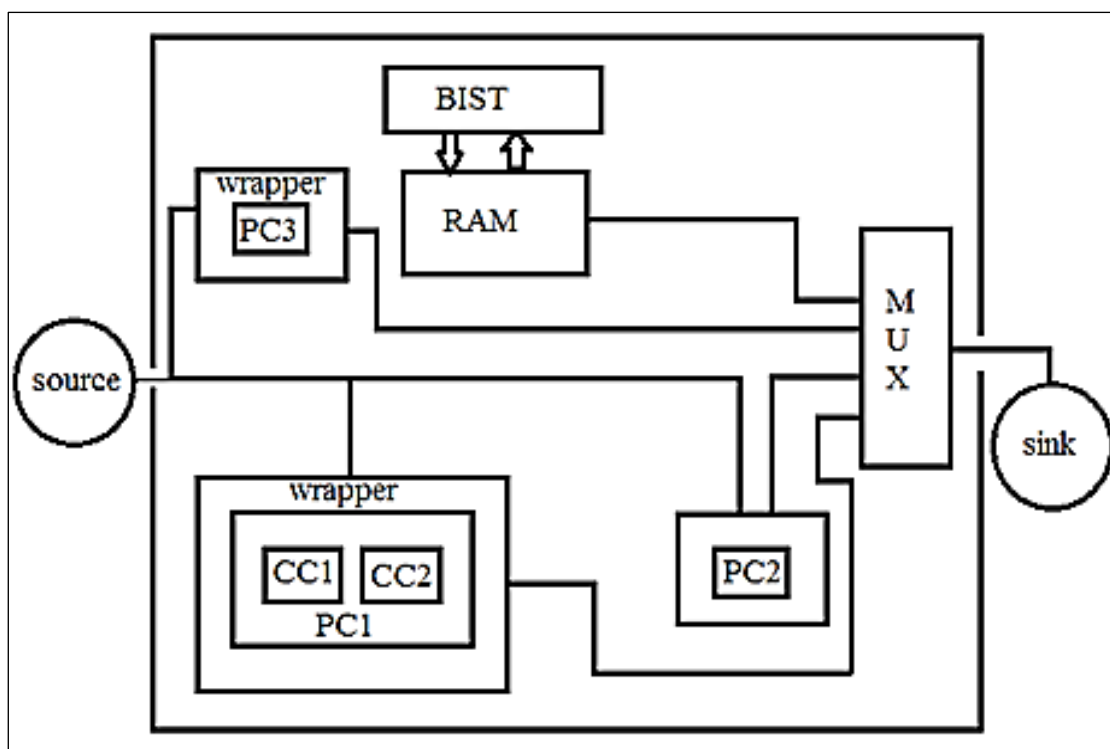


Fig. 1: SOC Test Model.

Large test data volume requires large test application time and hence SOC integrator has to decide on the concurrent scheduling of the testing of various cores using test access mechanism (TAM). TAM optimization and test scheduling at the SOC level have been an active area of research in recent years. Prior work has, however, been limited to traditional two-dimensional (2D) integrated circuits (ICs), which had been the widely used technology for IC manufacturing. With the shift to 3D structure the test planning decisions have been refined more to take into consideration the constraint of the through silicon vias. The cores lying in these 3D structures can be either 2D or 3D which necessitates the TAM optimization and the wrapper design to be modified to facilitate the changes.

TEST DATA VOLUME MINIMIZATION

With increase in the level of integration, SOC design technology (3D structures etc.) and working at deep submicron level, the possibility of occurrence of the manufacturing faults have increased. Processes like imperfect implantation, shorts between various interconnect, misalignment, thinning etc. lead to various faults which may affect the functionality and performance of the ICs. To ensure the quality against such faults, the test data has increased significantly. Meanwhile, the un-correlation between the successive test vectors aggravates the problem of the switching activity and hence the dynamic power dissipation.

In these years, the ATE have not advance at the same pace as their SOC counterparts in terms of the clock speed and on chip memory. The fact that only limited pins are available for carrying the test data necessities the transfer of the test bits to be done serially. Since the ATE frequency is comparatively small so the test vector delivery takes substantial amount of time. To save the available ATE memory and speed up the test vector transportation speed following techniques are being used.

Test Data Compression

The required test data can either be generated on chip using Built in self-test (BIST) or off chip using the Automatic test pattern generator (ATPG). In BIST approach, pseudo random

test patterns are generated by putting some deterministic seed in structures like linear feedback shift registers, ring counters, etc. However, to use the BIST structures the cores need to be BIST ready which may not feasible all the time as the cores are provided by third sources. Hence, the ATPG approaches are much better option wherein the test vectors are generated to achieve near 100 % fault coverage and saved in the memory of ATE.

Run Length Based Compression Techniques

Run length based compression techniques exploit the various combinations of test data to reduce the number of bits as much as possible. On analysis it is found that the test data dedicated for digital on chip testing consists of three types of data values: '0', '1' and 'X' (do not care). These values are present in various combinations like continuous runs of same value, unique (mix of 0's, 1's and X's) values and sandwiching X's between consecutive runs of 0's /1's. Many researchers have exploited such combinations to encode them using lesser number of bits termed as compression of test data. Such compressed patterns are saved in ATE memory. Such test data is transferred to the circuit under test where it is decoded using the on chip decompressor and applied to the circuit under test as shown in the Figure 2.

To decode, the code word is replaced back by the corresponding original uncompressed test data. The effectiveness of the various coding techniques are judged on the basis of the corresponding compression efficiencies can be calculated by using the uncompressed test length (T_D) and compressed test length (T_E) as per the equation 1.

$$\text{Compression efficiency} = \frac{\text{Uncompressed Test Length} - \text{Compressed Test Length}}{\text{Uncompressed Test Length}} * 100 \quad (1)$$

The successive test patterns in a test sequence often differ in only a small number of bits.

It has been observed that the test data consists of a large number of do not care bits and the successive test patterns normally differ in very few number of bits.

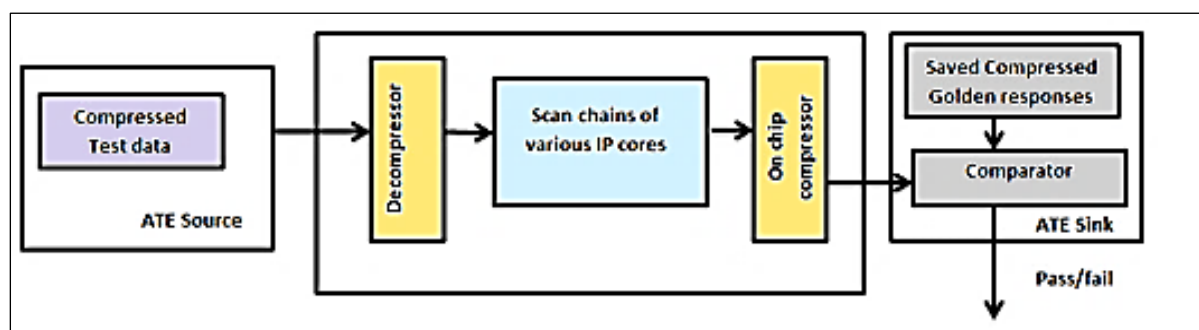


Fig. 2: Test Data Compression/Decompression.

This property is used in the many run length codes. The difference vector is generated by exclusive ORing the successive patterns. Since normally only few bits differ at various bit positions so the EX OR operation helps to obtain runs of zeroes in difference vectors. The Golomb Code [6] works on the compression by encoding the continuous runs of *all zeroes*. And the encoding is done on the basis of group prefix and tail selection. However, it offers inefficiency in the encoding if there exists only a small number of zeroes. For example, the size of the code generated for '0-4' runs of zeroes are encoded using the same number of bits which leads to expansion of bits instead of compression. This issue is removed in the FDR [7] code by making use of variable group sizes and hence the compression is increased. An extended FDR code [8] works on achievement of even better compression by encoding runs of *all zeroes* and *all ones*. These techniques provide a reasonably good amount of test data compression. Another technique that works on the encoding of the fixed length test data blocks using nine different codes [9]. The nine C compression code [10], considers the unique combinations along with the runs of zeroes and ones.

The blocks of fixed length on comparison can be encoded according to the existence of *all zeroes* (00), *all ones* (11), sub block of *all zero* followed by sub block of *all ones* (01), sub block of *all ones* followed by sub block of *all zero* (10), *all zero* followed by a sub block of *unique combination* (0U), *all ones* followed by a sub block of *unique combination* (1U), sub block of *unique combination* followed by *all zero* (U0), sub block of *unique combination* followed by *all ones* (U1), sub block of *unique combination* followed by another sub block of *unique*

combination (UU). Nine Coded compression technique is presented in two variants [10], fixed length 9 C wherein a block of fixed size (multiple of two) is used for encoding and variable 9c in which variable length two blocks are used for compression. However, only two blocks are used for compression which limits its compression efficiency.

Block merging codes (like BM, BM-8C [11], VPRL [12], CCPRL [13]) increase the efficiency by compressing Compatible blocks of fixed sizes. BM-8 C [11] integrates the compatibility based block merging and eight codes based intra block merging techniques. However, BM-8C treats compatible and inverse compatible blocks differently i.e. it allows blocks to be merged till compatibility is found. The moment an inverse compatible block is found, it is forwarded to next round of investigation for compatibility check. Also, the all zeroes and all ones patterns are not encoded efficiently. CCPRL and VPRL techniques perform Block merging to include both compatible and inverse compatible blocks. CCPRL technique also removes the unnecessary end bit used in VPRL. However, CCPRL has a limitation of adding extra bits (Count code) corresponding to representation of unique blocks Apart from test application time and ATE memory, test power is another major challenge faced by test engineers. Test power is normally very high as compared to power dissipated during the normal operation. The comparison of the test data compression achieved by various techniques as presented in Table 1. Various techniques have been developed to reduce the test vector scan in and scan out powers. Optimum do not care filling and vector reordering can be used to reduce the test power as well [14].

Table 1: Comparison of Compression Efficiencies (in percentage) Achieved between Proposed Compression Schemes and the Various Other Techniques.

Benchmark circuit	Golomb	FDR	EFDR	9C	B.M	2 ⁿ -PRL	B.M-8C	MS-HUFFMAN
S5378	37.11	48.02	53.67	51.64	54.98	54.94	58.56	61.2
S9234	42.25	43.6	48.66	50.94	51.19	57.72	57.49	61.7
S13207	79.74	81.3	82.49	82.31	84.89	88.1	87.52	81.5
S15850	62.82	66.22	68.66	66.38	69.49	64.29	73.69	71.4
S38417	28.37	43.26	62.02	60.63	59.39	58.33	59.92	67.5
S38584	57.17	60.93	64.28	65.53	66.86	72.44	71.66	70.9
Average	51.24	57.22	63.29	62.38	64.46	65.97	68.14	69.02

Statistical Codes and Dictionary Codes

The statistical coding techniques work on basis of the frequency of occurrence of same bit patterns. The patterns having higher frequency of occurrence are encoded using lesser number of bits (codes) as compared to once having lesser frequency. Huffman coding and its extended techniques like modified selective Huffman coding [15] are all examples of such statistical coding techniques. The results for M. S. Huffman coding as obtained for the various benchmark circuits are presented in column 9 of Table 1. The statistical techniques helps in achieving a good amount of average compression but the need of look up Table makes the decoding process test data dependent. So, based on the trade-offs between the compression requirement and the decoder area, one out of the several run length techniques can be opted. Another efficient technique based on efficient bit mask and dictionary selection is presented. N bit pattern are encoded with corresponding code-words based on dictionary. In this scheme, the uncompressed pre-generated test data is segmented into equal length blocks. Such blocks are further encoded by replacing it with a unique code kept as a corresponding dictionary entry. Since based in the data the number of such entries can be quite high so the associated overhead can be in feasible in maintenance.

Linear-Decomposition Based Schemes

Any decompressor consisting of only wires, XOR gates and flip flops is a linear decompressor [16] and has the property that its output space (the space of all possible vectors that it can generate) is a linear subspace spanned by a Boolean matrix. A linear decompressor can generate test vector Y if and

only if there exists a solution to the system of linear equations $AX = Y$, where A is the characteristic matrix for the linear decompressor and X is a set of free variables shifted in from the tester (you can think of every bit on the tester as a free variable assigned as either 0 or 1). The characteristic matrix for a linear decompressor is obtainable from symbolic simulation of the linear decompressor. Two types of linear decompression techniques have been proposed.

1. *Combinational Linear Decompression:* Combinational linear decompression technique employs each scan chain to be fed by the value generated by XOR-ing some small set of free variables (equal to the number of channels) coming from the tester [17, 18]. That makes the hardware and its control to be relatively simple. To achieve high compression efficiency, it needs the number of channels to be sufficiently large to encode the most highly specified scan slices which is quite limited in average number of cases.

2. *Sequential Linear Decompression:* In this the decompressor is based on linear finite-state machines such as LFSRs, cellular automata, or ring generators [19, 20]. Their advantage lies in allowing the use of free variables from earlier clock cycles to encode a scan slice in the current clock cycle. This provides much greater flexibility than combinational decompressors. However the decompressors are more complex. Sequential linear decompression can be differentiated into of two types based on initial seeding: static reseeding and dynamic reseeding.

Broadcast-scan-based Schemes

In this scheme same value is broadcasted to multiple scan chains [21, 22]. It is a special

degenerate case of linear decompression in which the decompressor consists of only fan-out wires. The dependent inputs are tied together in the circuit description given to the ATPG so that the ATPG algorithm will produce only encodable test cubes. It is not easy to do this for linear decompressors that use XORs, because incorporating XOR gates can significantly degrade ATPG performance. Reconfiguring the sets of scan chains that each tester channel broadcasts to, can increase the number of faults detected by the broadcast-scan based schemes.

TEST ARCHITECTURE DEVELOPMENT

As explained in the Section 2.1, the test architecture development plays a bigger role in deciding the overall test cost associated with the SOC. The problem of the test architecture development can be categorized into a number of steps namely: test wrapper development; test access mechanism design and test scheduling. These are as explained below.

Wrapper Design

The core test wrapper design is an important part in deciding the test time of a core T to the available TAM width. It helps in stitching the various functional input/output cells and internal scan chains of the circuit under test/core to match the available TAM. The total test time corresponding to the available TAM can be calculated as per the formula given in equation 2.

$$T_i = (1 + \max(S_i, S_o) p) + \min(S_i, S_o) \quad (2)$$

Where T_i refers to the core test time and S_i , S_o refers to the input and output scan chain lengths.

The wrapper design algorithm works in the balancing of the internal scan chains such that the difference between the maximum input and output scan chain lengths is minimum. To facilitate the testing of the components inside and around the core and operation of the core in the normal mode, a test wrapper can work in following mode as described [23]:

- Internal test mode: allows the testing of internal components of the circuit under test.

- External mode that allows the testing of the interconnections between different cores.
- Bypass mode to bypass the cores which need not be tested during that very moment. Basically it is required to avoid the unnecessary switching associated with the terminals of the core which is not being tested.
- Normal mode that allows the normal execution of the core.

In an SOC the cores may be in either wrapped or unwrapped form. A P1500 core wrapper shown in Figure 3 supports both core-internal and core external testing through the use of INTEST and EXTEST modes [24, 25]. Another wrapper approach, called Test Collar, is proposed by Verma and Bhatia [25]. The approach is similar to the test shell; however, the approach does not have the bypass feature which reduces the flexibility by allowing only one core to be served at a time. An approach combining P1500 and the Test Shell approach has been proposed by Marinissen *et al.* A major advantage is the flexible bypass introduced. Two types of bypass styles are defined; wrapper bypass and scan -chain bypass. The wrapper bypass is the same as used in the test shell while the scan -chain bypass is flexible structures which can be inserted at any place between a terminal input and a terminal output. It allows a non-clocked bypass structure which can be used to bypass the complete core. Reconfigurable Core Wrappers proposed by Koranne [26] allows, in contrast to approaches such as Boundary Scan, Test Shell and P1500, with several wrapper chain configurations the flexibility is increased in the scheduling process. For each TAM widths a Di-graph (directed graph) is generated where a node denotes a scan -chain and the input TAM. K. Chakarbarty *et al.* formulated the wrapper design problem in terms of breadth first and bin packing algorithm while abiding by the P1500 standard. All the wrapper design algorithms developed till 2008 majorly concentrated on the Flat SOCs. However in real world there are heterogeneous cores comprising of hierarchical cores which demands a separate test access for parent and child cores.

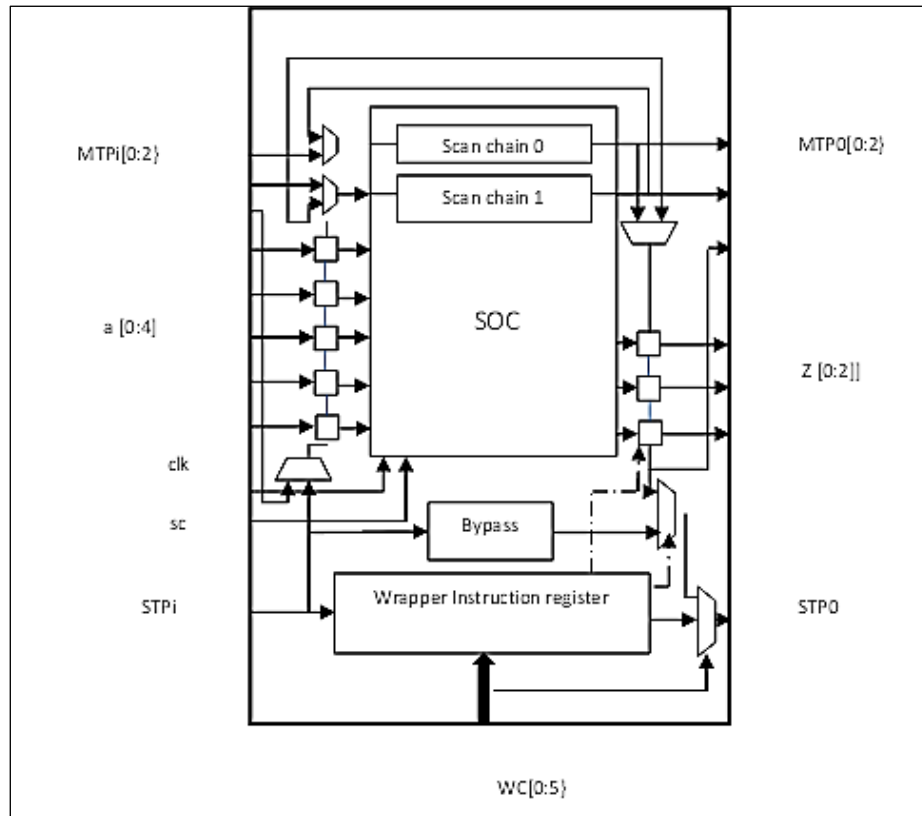


Fig. 3: IEEE P1500 Wrapper Architecture.

All the wrapper designs Goel *et al.* [27] and Kyuchull Kim *et al.* [28] have proposed a wrapper design for the Hierarchical SOC's. These perform the wrapper cells design by accepting the available TAM count and then try to balance the internal wrapper chains for both parent and child. It does not only allow parallel testing of a parent core and its child core(s) but also helps determining the requirements of parent and child EXTEST and INTTEST modes. With the expansion in the vertical dimension, 3D SOC's are gaining considerable attention by the designers. To ensure the testing, modular test approach which was applicable for the 2D SOC's have to be modified to cover the various constraints specific to such systems. The test planning approach so formed should ensure to take care of the limits set by Through Silicon Vias (TSV), power, core placement etc. based on the designing of the 3D SOC's, these can be categorized as fine grain partitioned 3D SOC's and course grained partitioned 3DSOC's. The fine grained SOC's refer to the structures wherein the cores spread over multiple layers while in course grained, the cores are still 2D but their placement is 3D. Therefore, corresponding to the different core formation,

the wrapper designs have to be modified in fine grained. A core wrapper design that addresses the TSV constraint in addition to available TAM is proposed. Once the wrappers have been properly designed, any 2D test assignment and schedule can be followed to reduce the test time. One thing that needs to be taken care of is that all the core terminals start and end at ground die which means all the test wires that go up have to come down which affect the number of the TSV's used. Scenario is different for course grained partition SOC's wherein, major change which needs to be done is in the TAM design. Since the core design is 2D so the 2D wrapper design is still applicable. However, the TAM wires need to be routed between the different dies to allow testing. One such approach is proposed by Wu *et al.* [29]. A structured and scalable DFT base modular test approach is proposed in [20] that allow pre, partial and post bond testing. The dies and their associated cores are tested as separate units. The approach uses the available test architectures of the dies/ core dedicated probe pads, test elevators (to move the test data and instructions between the dies) and hierarchical wrapper instruction registers chains.

Test Access Mechanism

A major difficulty concern is accessibility of embedded cores from the I/O terminals of the system. Since the number of pins is always a limitation so the test bandwidth available on chip has to be used judiciously to deliver test patterns to the individual core and observing core responses [30, 31]. The data is carried by the test access mechanism between the ATE and the pins of various IP cores as shown in (Figure 4). The approaches can be either to multiplex the core terminals to chip level pins so that the test patterns can be directly applied and observed (which simplifies the TAM problem but is not much of use due to wiring overhead owing to increase in number of cores and the difference between the total number of the cores pins and the overall pin limitation at the SOC side).

Aerts and Marinissen [32] described three different bus (shown in Figure 5) based TAMs: Multiplexed architecture gives each core access to the full TAM width on time multiplexed bases (shown in Figure 5 (a))

leading to longer test time. The second architecture proposed [32] is called Daisy chain as shown in Figure 5(b). It connects all cores through one long TAM. In this technique, each core is allotted whole test data width which leads each core to be tested in minimum possible time. Each core is parallel connected to a bypass cell too which helps in isolation of the cores for the test purpose. Once a specific core has been tested then it can be bypassed during the test of the successive cores so as to reduce the test time. The third approach is the distribution approach as shown in the Figure 5(c). Using this technique the complete TAM is divided in partitions of varying TAM widths. The cores assigned to different TAM partitions can be tested concurrently irrespective of previous two TAM schemes wherein the testing of the cores need to be done sequentially. The major drawback of the distribution TAM architecture over others is that it does not allow the interconnection testing that lie between the different cores.

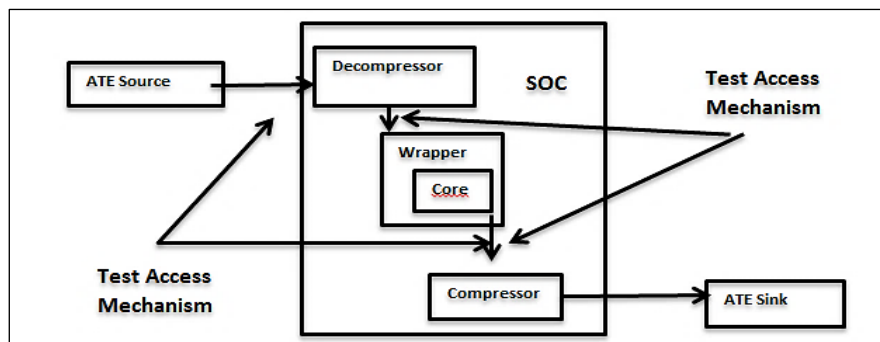


Fig. 4: Modular Test Approach of SOC.

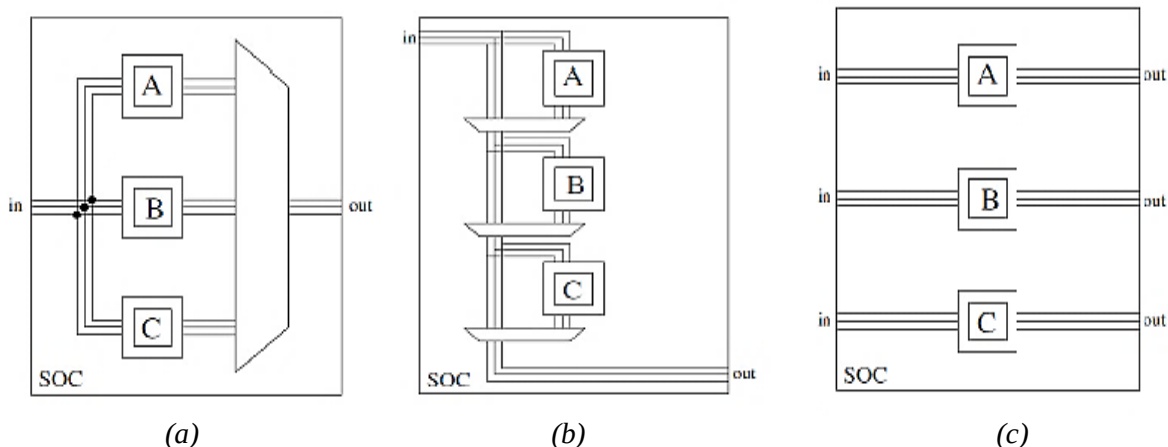


Fig. 5: Test Architecture Designs; (a) Multiplexing Architecture; (b) Daisy Chain; (c) Distributed Architecture.

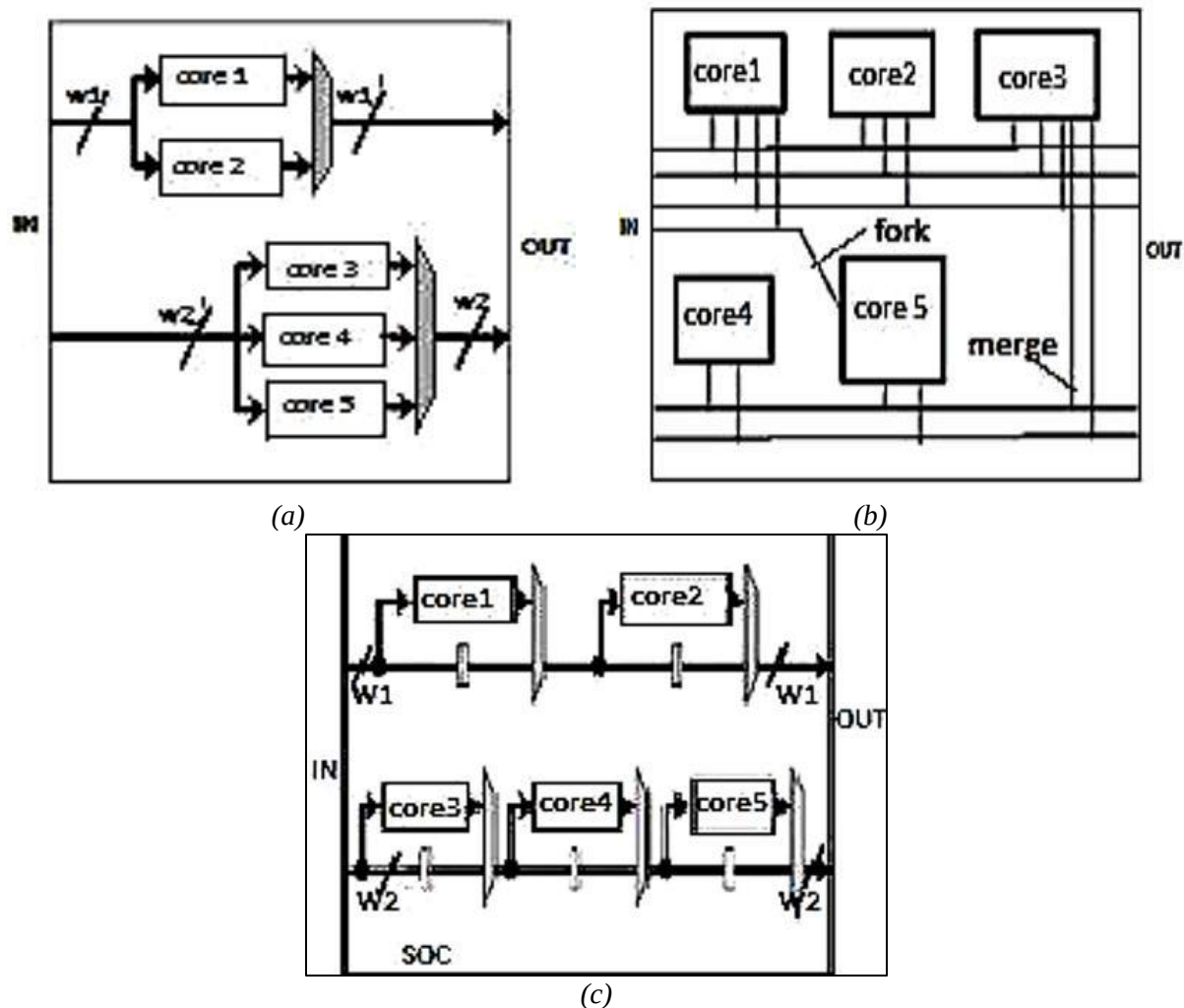


Fig. 6: TAM Types: (a) Test Bus; (b) Flexible TAM with Fork and Merge; (c) Testrail.

A combination of the multiplexed and distributed TAM approaches is the test bus approach [25] as shown in Figure 6(a). As pointed by the Figure, this type of approach offers both the sequential and concurrent testing. Cores allotted to the same TAM have to be tested sequentially while the cores assigned to different TAM partitions can be tested concurrently. Two variants of the test bus approach [21] are the fixed width test bus (Figure 6(a)) and flexible width test bus (Figure 6(b)). In case of the flexible width test bus more flexibility is given to further partition and test buses on basis of improvement in the test time. However, due to partitioning still the interconnection testability may suffer. An amalgamation of the daisy chain and the distribution architectures is test rail architecture as shown in the Figure 6(c). It allows all the cores to be allotted the maximum available TAM width as per the

assigned partition [33]. The bypass path in the individual test buses is same as that of the daisy chain architecture.

The width of a testrail is referred to as the test data width, since it determines the overall system testing time. Cores connected to a TAM of type test rail can be tested simultaneously or sequentially. A simultaneous test of two cores can be done by concatenating the scan chains and wrappers of the cores. All the above mentioned techniques assumed the SOC to be flat with cores. But later on it was found that the different requirements of the parent and child cores in the hierarchical SOC necessitate the need to look at the problem differently. Iyengar *et al.* [34] shown in Figure 7(a) has presented a multilevel TAM optimization technique for TAM width adaptation using a combination of integer linear programming (ILP) and

enumeration for both the interactive and non-interactive types of cores provided by different vendors with a perspective to reduce test time. Goel *et al.* [35] proved how the efficient design of the TAM helps reduce the unnecessary addition of extra bits though proper distribution of the TAM lines.

The most recent advancements in the 3D technology gave rise to another consideration parameter which is to do with the number of through silicon vias to come up as another limitation on TAM design. Xiaoxia Wu *et al.*

[36] have extended the TAM design problem based upon ILP formulation for 3D SOCs. A conceptual view of the 3D TAM is as shown in Figure 7(b). The cores are assumed to be non-partitioned i.e non 3D however they are assumed to lie on different layers. The number of the dies is taken to be 3 along with number of TAM (N_{TAM}) being 3. Marinissen *et al* has developed a structured and scalable test access architecture for TSV-based 3D stacked ICs for a pre and post bond testing. It incorporates the IEEE 1149.1 and IEEE 1500 wrapper standard.

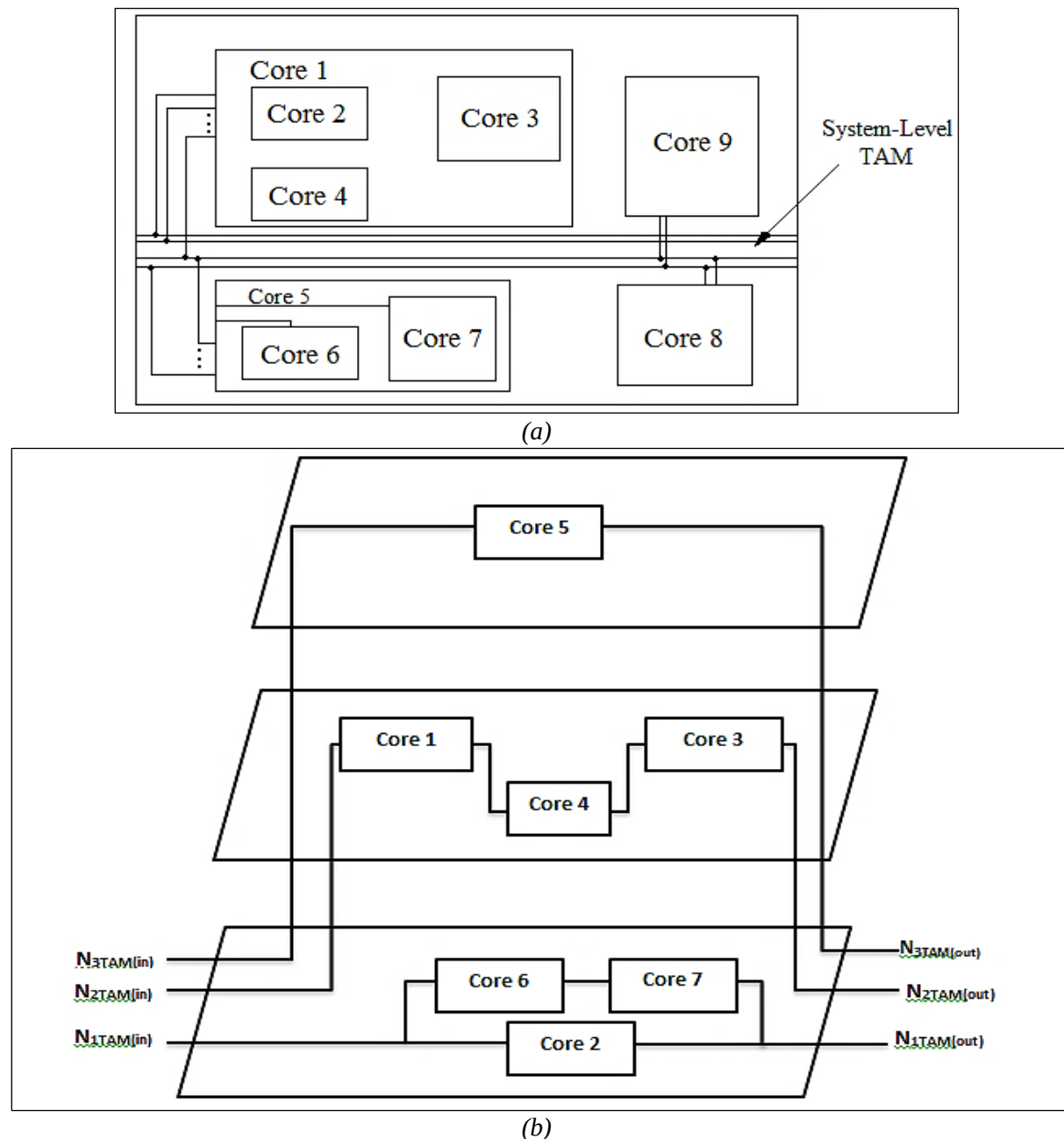


Fig. 7: (a) TAM Design for Hierarchical Cores (b) TAM Design for 3D SOC [30].

Test Scheduling

The schedule of the core testing basically aims at of defining the start and end time of the testing of various cores. The problem of the test assignment and scheduling has been proved to be NP hard problem [34, 35]. As stated, the testing of the cores assigned to the TAM can either be done sequentially or concurrently. In sequential test scheduling, all tests are ordered in a sequence where only one test is applied at a time, while in concurrent test scheduling more than a single test can but does not have to be active at a time. To reduce the testing time associated with the application of the test data to various cores, the testing of cores should be done as concurrently as possible. However the extent of concurrency is often limited by the constraints set by the test power, resources (BIST, core wrapper boundary terminals, precedence and hierarchy. Testing of the cores can be done concurrently to reduce the test access time.

Test scheduling techniques can be divided into 3 types:

- Session-based testing
- Session less testing with run to completion.
- Pre-emptive testing, in which a core's test can be interrupted and resumed later.

To do the concurrent testing, cores are grouped based on their compatibility with each other. Two cores are said to be compatible if they don't have any resource, hierarchy, power etc conflicts with respect to each other. In session based testing approach, compatible cores are scheduled in terms of sessions. The length of the session is chosen to be the maximum of testing time of individual cores. Every session based test scheduling technique results in long test application time. This is because idle times are generated when the test schedule is divided into sessions. Hence, the other two session less test scheduling schemes are used.

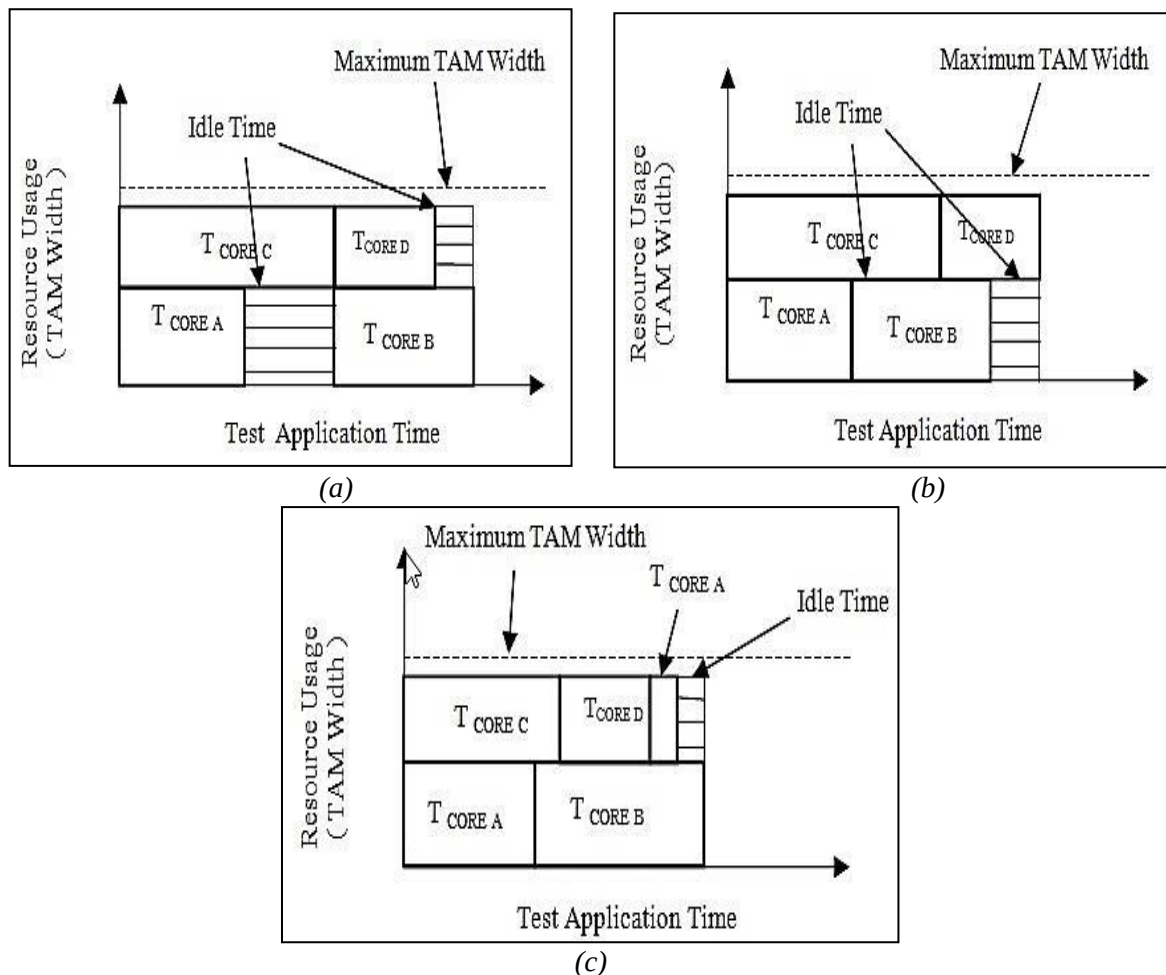


Fig. 8: Test Scheduling Techniques: (a) Non-partitioned (b) Partitioned (c) Pre-emptive.

In session less scheduling, an attempt is made to remove the drawback of the idle time. New cores are scheduled as soon as the core assigned to the TAM partition finishes its testing time. Therefore, the idle times are taken to the end of the schedules. An example of such schedule is shown in Figures 8(a and b). Another variant of the sessionless schedule is pre-emptive scheduling. In this type of schedule the problem of the idle time reduction is improved even more by employing pre-emption technique. Herein, the test of the cores are halted in between and restarted later on based on the availability of the TAM. An example of such technique is shown in Figure 8 (c). However this kind of scheduling is not a very good practical. For example: the test vectors of sequential circuits are dependent on each other therefore if continuity is broken, it can jeopardise the very purpose of testing. Similarly, memory BIST operations can also not be pre-empted [37]. Best approach is to sort out the test compatible cores and schedule them allowing their complete test.

A lot of effort has been paid on test scheduling and new better algorithms have been purposed like genetic, simulated annealing based, ILP, PSO, Job shop scheduling. A hierarchy aware test schedules [27] have also been developed that takes care of the precedence constraint among the child and parent cores while abiding by various constraints.

CONCLUSIONS

This paper provided a brief perspective on modular test approach and the challenges in testing of cores in System-on-chip. The challenges related to minimizing test cost function which depends on the testing time has been briefly illustrated. The solution to it is adequate test planning which includes test data volume reduction and test access architecture optimization has been presented by the authors. On the basis of the vast literature survey it can be observed that the test data volume reduction techniques still offers an options for the further research so as to reduce the ATE Memory, power and time constraints. Also, TDV reduction techniques can be merged with the low power scan based techniques to further reduce the power. Same

can be done while taking the hierarchical cores 3D SOC test optimization needs to be done with respect to the power constraint. So in simple words SOC test offers a vast options for new technological innovations to address the challenges predicted by the ITRS roadmap.

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AUTHORS' BIBLIOGRAPHIES



Harpreet Vohra received her Master in VLSI Design. She joined Thapar University in year 2006 and is working as Assistant Professor with Electronics and Communication Engineering Department. Her Interest areas include low

power VLSI design and test. She has guided around 14 M.Tech. Thesis.



Amardeep Singh received his B.Tech. degree in Electronics and Communication from MIT and his M.Tech degree in Computer Science and Engineering from Punjabi University, Patiala. He received his PhD degree in VLSI Testing using DNA and Quantum computing Algorithms in 2006 from Thapar University, Patiala. Now he is holding an academic position as Professor in Computer Engineering Department, Punjabi University, Patiala, India. His main interest areas include nanocomputing, embedded systems, and nonconventional algorithms.

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